

NSF 25-543: Computer and Information Science and Engineering : Future Computing Research (Future CoRe)

Program Solicitation

Document Information

Document History

- **Posted:** July 25, 2025
- **Replaces:** [NSF 24-589](#), [NSF 24-581](#), [NSF 25-527](#)

[View the program page](#)



U.S. National Science Foundation

Directorate for Computer and Information Science and Engineering

Division of Computing and Communication Foundations

Division of Information and Intelligent Systems

Division of Computer and Network Systems

Full Proposal Target Date(s):

September 11, 2025

Second Thursday in September, Annually Thereafter

February 05, 2026

First Thursday in February, Annually Thereafter

While proposals are accepted anytime, proposers are highly encouraged to submit by the target dates to ensure consideration during the corresponding panel review cycle.



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Important Information And Revision Notes

- Added Computing Education Research (CER) and Cyber-Physical Systems Foundations and Connected Communities (CPS) programs.
- There are no deadlines for all proposals, but target dates have been added to enhance the timing of review.
- Replaced Medium and Small project classes with a single project class with a maximum budget of up to \$1,000,000 and a duration of up to 4 years.
- Removed the OAC Core program; proposers are encouraged to submit to the closest Future CoRe program instead.
- Modified solicitation-specific eligibility criteria for organizations and PIs.
- Removed solicitation-specific review criteria, award conditions, and reporting requirements.
- Removed requirement for collaboration and broadening participation in computing plans.

Any proposal submitted in response to this solicitation should be submitted in accordance with the *NSF Proposal & Award Policies & Procedures Guide* (PAPPG) that is in effect for the relevant due date to which the proposal is being submitted. The NSF PAPPG is regularly revised and it is the responsibility of the proposer to ensure that the proposal meets the requirements specified in this solicitation and the applicable version of the PAPPG. Submitting a proposal prior to a specified deadline does not negate this requirement.

Summary Of Program Requirements

General Information

Program Title:

Computer and Information Science and Engineering (CISE): Future Computing Research (Future CoRe)

Synopsis of Program:

The NSF CISE Directorate supports research and education projects that develop new knowledge in all aspects of computing, communications, and information science and engineering through the following Future Computing Research (Future CoRe) programs:

- Algorithmic Foundations (AF) program;
- Communications and Information Foundations (CIF) program;
- Computer Systems Research (CSR) program;
- Computing Education Research (CER) program;
- Cyber-Physical System Foundations and Connected Communities (CPS) program;
- Foundations of Emerging Technologies (FET) program;
- Human-Centered Computing (HCC) program;
- Information Integration and Informatics (III) program;
- Networking Technology and Systems (NeTS) program;
- Robust Intelligence (RI) program; and
- Software and Hardware Foundations (SHF) program;

The CISE Future Computing Research program anticipates a portfolio of awards with a range of budgets and durations, including projects of smaller scope. Project durations and budgets must be commensurate with the scope of the proposed work up to the maximum limit of \$1,000,000 with a duration up to 4 years. Typical projects are approximately \$150,000 to \$250,000 per year and are 3 to 4 years in duration. Projects are discouraged from exceeding \$300,000 in any single year. Estimated program budget, number of awards, and average award size/duration are subject to the availability of funds.

Broadening Participation In STEM:

NSF has a mandate to broaden participation in science and engineering, as articulated and reaffirmed in law since 1950. Congress has charged NSF to "develop intellectual capital, both people and ideas, with particular emphasis on groups and regions that traditionally have not participated fully in science, mathematics, and engineering."

Cognizant Program Officer(s):

Please note that the following information is current at the time of publishing. See program website for any updates to the points of contact.

- AF Program Team, telephone: (703) 292 8910, email: cise-af@nsf.gov
- CIF Program Team, telephone: (703) 292 8910, email: cise-cif@nsf.gov

- CSR Program Team, telephone: (703) 292 8950, email: cise-csr@nsf.gov
- CER Program Team, telephone: (703) 292 8910, email: cise-cer@nsf.gov
- CPS Program Team, telephone: (703) 292 8950, email: cise-cps@nsf.gov
- FET Program Team, telephone: (703) 292 8910, email: cise-fet@nsf.gov
- HCC Program Team, telephone: (703) 292 8930, email: cise-hcc@nsf.gov
- III Program Team, telephone: (703) 292 8930, email: cise-iii@nsf.gov
- NeTS Program Team, telephone: (703) 292 8950, email: cise-nets@nsf.gov
- RI Program Team, telephone: (703) 292 8930, email: cise-ri@nsf.gov
- SHF Program Team, telephone: (703) 292 8910, email: cise-shf@nsf.gov

Applicable Catalog of Federal Domestic Assistance (CFDA) Number(s):

- 47.070 --- Computer and Information Science and Engineering

Award Information

Anticipated Type of Award: Standard Grant or Continuing Grant

Estimated Number of Awards: 400 to 600

Anticipated Funding Amount: \$280,000,000

Dependent upon the availability of funds.

Eligibility Information

Who May Submit Proposals:

Proposals may only be submitted by the following:

- Institutions of Higher Education (IHEs): Two- and four-year IHEs (including community colleges) accredited in, and having a campus located in the US, acting on behalf of their faculty members. Special Instructions for International Branch Campuses of US IHEs: If the proposal includes funding to be provided to an international branch campus of a US institution of higher education (including through use of sub-awards and consultant arrangements), the proposer must explain the benefit(s) to the project of performance at the international branch campus, and justify why the project activities cannot be performed at the US campus.
- Non-profit, non-academic organizations: Independent museums, observatories, research laboratories, professional societies and similar organizations located in the U.S. that are directly associated with educational or research activities.
- Tribal Nations: An American Indian or Alaska Native tribe, band, nation, pueblo, village, or community that the Secretary of the Interior acknowledges as a federally recognized tribe pursuant to the Federally Recognized Indian Tribe List Act of 1994, 25 U.S.C. §§ 5130-5131.

Who May Serve as PI:

There are no restrictions or limits.

Limit on Number of Proposals per Organization:

There are no restrictions or limits.

Limit on Number of Proposals per PI or co-PI: 2

For CISE Future CoRe Programs, an individual may not serve as a PI, co-PI, or Senior/Key Personnel on more than two proposals submitted within any consecutive 12-month period across all Future CoRe programs listed in this solicitation.

Proposals submitted to the previous version of CISE core Programs (NSF 24-589); Cyber-Physical Systems (NSF 24-581); Smart and Connected Communities (NSF 25-527) or any other NSF funding opportunities do not count towards the limit on the number of proposals per PI, co-PI or Senior Personnel.

These eligibility constraints will be strictly enforced to treat everyone fairly and consistently. Any proposal that exceeds this limit at the time of submission for any PI, co-PI, or Senior/Key Personnel will be returned without review. **No exceptions will be made.** Proposals that are withdrawn prior to commencement of the merit review process, or those that are returned without review by NSF, will not count against this proposal limit. Proposers are strongly encouraged to verify the dates of prior submissions to Future CoRe Programs for all personnel on their teams to avoid their proposals being returned without review.

Proposal Preparation and Submission Instructions

A. Proposal Preparation Instructions

- **Letters of Intent:** Not required
- **Preliminary Proposal Submission:** Not required
- **Full Proposals:**
 - Full Proposals submitted via Research.gov: *NSF Proposal and Award Policies and Procedures Guide* (PAPPG) guidelines apply. The complete text of the PAPPG is available electronically on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.
 - Full Proposals submitted via Grants.gov: *NSF Grants.gov Application Guide: A Guide for the Preparation and Submission of NSF Applications via Grants.gov* guidelines apply (Note: The *NSF Grants.gov Application Guide* is available on the Grants.gov website and on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=grantsgovguide).

B. Budgetary Information

- **Cost Sharing Requirements:**

Inclusion of voluntary committed cost sharing is prohibited.
- **Indirect Cost (F&A) Limitations:**

Not Applicable
- **Other Budgetary Limitations:**

Not Applicable

C. Due Dates

- **Full Proposal Target Date(s):**

September 11, 2025

Second Thursday in September, Annually Thereafter

February 05, 2026

First Thursday in February, Annually Thereafter

While proposals are accepted anytime, proposers are highly encouraged to submit by the target dates to ensure consideration during the corresponding panel review cycle.

Proposal Review Information Criteria

Merit Review Criteria:

National Science Board approved criteria apply.

Award Administration Information

Award Conditions:

Standard NSF award conditions apply.

Reporting Requirements:

Standard NSF reporting requirements apply.

I. Introduction

The NSF Directorate for Computer and Information Science and Engineering (CISE) supports transformative research and education projects that develop new knowledge in all aspects of computing, communications, and information science and engineering through multiple research programs. These programs support research and education activities that advance:

- mathematical, scientific and technological foundations of computing communication, hardware, software and emerging technologies;
- understanding and development of computer and network systems, cyber-physical systems, and cybersecurity as well as their roles in solving complex scientific, engineering, and societal problems; and
- understanding of the inter-related roles of people, computers, and information.

II. Program Description

This solicitation covers submission to the following Future CoRe programs. Research that fits within a single program and interdisciplinary research that spans more than one of these programs are welcome.

Please see the individual program webpages below for more information on what is within scope for these Future CoRe programs:

- Algorithmic Foundations (AF) program [[Program Webpage](#)] supports potentially transformative projects in the theory of algorithms and computational complexity, characterized by algorithmic innovation and rigorous analysis;
- Communications and Information Foundations (CIF) program [[Program Webpage](#)] supports foundational research that addresses the theoretical underpinnings of information acquisition, transmission, and processing in communications and information processing systems;
- Computer Systems Research (CSR) program [[Program Webpage](#)] supports the advancement and holistic design and development of integrated software and hardware computing systems, including classical and quantum elements, to enable an integrated, cohesive, intelligent computational ecosystem;
- Computing Education Research (CER) program [[Program Webpage](#)] supports transformative projects that inform our understanding of the effective teaching and learning of computing skills and concepts at all levels to contribute to the development of, and pathways toward a robust workforce well prepared for careers in CISE fields;

- Cyber-Physical Systems Foundations and Connected Communities (CPS) program [[Program Webpage](#)] supports convergent research on intelligent, engineered systems, built from the seamless integration of computation with physical components within natural and built environments. The program seeks foundational and translational advances in all areas of cyber-physical systems, including artificial intelligence for cyber-physical systems, and encourages multidisciplinary socio-technical innovation, transforming the way people, technologies, and the environment interact in communities;
- Foundations of Emerging Technologies (FET) program [[Program Webpage](#)] supports foundational research at the intersection of computing and biological systems, nanoscale science and engineering, quantum information science, and other promising disruptive technologies supporting novel computing/communication models;
- Human-Centered Computing (HCC) program [[Program Webpage](#)] supports research in human-computer interaction, integrating across fields including computing, information, social and behavioral sciences, to (re)design technologies that amplify human capabilities, and understand how human, technical, and contextual aspects of computing and communication systems shape their benefits, effects, and risks;
- Information Integration and Informatics (III) program [[Program Webpage](#)] supports research on computational approaches to the full data lifecycle to maximize the utility of information resources;
- Networking Technology and Systems (NeTS) program [[Program Webpage](#)] supports research that advances intelligent communication network systems, including wired, wireless, Internet, internet of things, quantum, bio-inspired, nano, and other networks, as well as networks that span multiple domains and scales like space-air-ground integrated networks, while exploring their fundamental understanding, properties and trade-offs, and developing innovative tools and techniques, including artificial intelligence (AI), for designing, building, measuring and managing future communication network systems and services;
- Robust Intelligence (RI) program [[Program Webpage](#)] supports computational research to understand and enable intelligent systems in complex, realistic contexts; and
- Software and Hardware Foundations (SHF) program [[Program Webpage](#)] supports foundational research in the design, verification, operation, utilization and evaluation of computer hardware and software through novel approaches, robust theories, high-leverage tools, and lasting principles.

The CISE Future Computing Research program anticipates a portfolio of awards with a range of budgets and durations, including projects of smaller scope. Project durations and budgets must be commensurate with the scope of the proposed work up to the maximum limit of \$1,000,000 with the duration up to 4 years. Typical projects are approximately \$150,000 to \$250,000 per year and are 3 to 4 years in duration. Projects are discouraged from exceeding \$300,000 in any single year. Estimated program budget, number of awards, and average award size/duration are subject to the availability of funds.

Expanding Geographic and Institutional Range in Computer and Information Science and Engineering

NSF CISE encourages proposal submissions from EPSCoR-eligible institutions to the CISE Future Computing Research programs, with an aim to enhance engagement within the science, technology, engineering, and mathematics (STEM) enterprise, specifically associated with geographic location, and thereby enabling the jurisdiction's national competitiveness. Through this initiative, CISE aims to promote funded activities that enable sustainable growth in research enterprise in EPSCoR jurisdictions. Collaborative proposals among the EPSCoR and Non-EPSCoR-eligible jurisdictions that are led by EPSCoR institutions are particularly welcomed.

Proposals for Consideration by Multiple CISE Future Computing Research Programs

For proposals that intersect with more than one CISE Future Research program, PIs must identify the most relevant programs in the proposal submission process. For information about submission and how to identify such proposals, refer to the Proposal Preparation Instructions section of this document. In these cases, PIs should also ensure that their proposals meet the program-specific guidelines for each identified research program. NSF CISE program officers will consider co-reviewing these proposals as appropriate.

III. Award Information

Up to \$280 million each year will support up to 600 awards, pending the availability of funds.

Estimated program budget, number of awards and average award size/duration are subject to the availability of funds.

IV. Eligibility Information

Who May Submit Proposals:

Proposals may only be submitted by the following:

- Institutions of Higher Education (IHEs): Two- and four-year IHEs (including community colleges) accredited in, and having a campus located in the US, acting on behalf of their faculty members. Special Instructions for International Branch Campuses of US IHEs: If the proposal includes funding to be provided to an international branch campus of a US institution of higher education (including through use of sub-awards and consultant arrangements), the proposer must explain the benefit(s) to the project of performance at the international branch campus, and justify why the project activities cannot be performed at the US campus.
- Non-profit, non-academic organizations: Independent museums, observatories, research laboratories, professional societies and similar organizations located in the U.S. that are directly associated with educational or research activities.
- Tribal Nations: An American Indian or Alaska Native tribe, band, nation, pueblo, village, or community that the Secretary of the Interior acknowledges as a federally recognized tribe pursuant to the Federally Recognized Indian Tribe List Act of 1994, 25 U.S.C. §§ 5130-5131.

Who May Serve as PI:

There are no restrictions or limits.

Limit on Number of Proposals per Organization:

There are no restrictions or limits.

Limit on Number of Proposals per PI or co-PI: 2

For CISE Future CoRe Programs, an individual may not serve as a PI, co-PI, or Senior/Key Personnel on more than two proposals submitted within any consecutive 12-month period across all Future CoRe programs listed in this solicitation.

Proposals submitted to the previous version of CISE core Programs (NSF 24-589); Cyber-Physical Systems (NSF 24-581); Smart and Connected Communities (NSF 25-527) or any other NSF funding opportunities do not count towards the limit on the number of proposals per PI, co-PI or Senior Personnel.

These eligibility constraints will be strictly enforced to treat everyone fairly and consistently. Any proposal that exceeds this limit at the time of submission for any PI, co-PI, or Senior/Key Personnel will be returned without review. **No exceptions will be made.** Proposals that are withdrawn prior to commencement of the merit review process, or those that are returned without review by NSF, will not count against this proposal limit. Proposers are strongly encouraged to verify the dates of prior submissions to Future CoRe Programs for all personnel on their teams to avoid their proposals being returned without review.

V. Proposal Preparation And Submission Instructions

A. Proposal Preparation Instructions

Full Proposal Preparation Instructions: Proposers may opt to submit proposals in response to this Program Solicitation via Research.gov or Grants.gov.

- Full Proposals submitted via Research.gov: Proposals submitted in response to this program solicitation should be prepared and submitted in accordance with the general guidelines contained in the *NSF Proposal and Award Policies and Procedures Guide* (PAPPG). The complete text of the PAPPG is available electronically on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg. Paper copies of the PAPPG may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov. The Prepare New Proposal setup will prompt you for the program solicitation number.
- Full proposals submitted via Grants.gov: Proposals submitted in response to this program solicitation via Grants.gov should be prepared and submitted in accordance with the *NSF Grants.gov Application Guide: A Guide for the Preparation and Submission of NSF Applications via Grants.gov*. The complete text of the *NSF Grants.gov Application Guide* is available on the Grants.gov website and on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=grantsgovguide. To obtain copies of the Application Guide and Application Forms Package, click on the Apply tab on the Grants.gov site, then click on the Apply Step 1: Download a Grant Application Package and Application Instructions link and enter the funding opportunity number, (the program solicitation number without the NSF prefix) and press the Download Package button. Paper copies of the Grants.gov Application Guide also may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov.

In determining which method to utilize in the electronic preparation and submission of the proposal, please note the following:

Collaborative Proposals. All collaborative proposals submitted as separate submissions from multiple organizations must be submitted via Research.gov. PAPPG Chapter II.E.3 provides additional information on collaborative proposals.

See PAPPG Chapter II.D.2 for guidance on the required sections of a full research proposal submitted to NSF. Please note that the proposal preparation instructions provided in this program solicitation may deviate from the PAPPG instructions.

Follow all the Proposal Preparation instructions in the PAPPG, plus these additional instructions:

Proposal Titles:

Proposal titles should begin with an acronym that indicates the most relevant Future CoRe program. Select a bolded acronym from the following list determined by the most relevant program area (if more than one area is relevant, please see instructions below):

- Algorithmic Foundations (**AF**) program;
- Communications and Information Foundations (**CIF**) program;
- Computer Systems Research (**CSR**) program;
- Computing Education Research (**CER**) program;
- Cyber-Physical Systems Foundations and Connected Communities (CPS) program (either **CPS-FR** for the Foundational Research in CPS track or **CPS-CIR** for the Community-Inspired CPS Research track);
- Foundations of Emerging Technologies (**FET**) program;
- Human-Centered Computing (**HCC**) program;
- Information Integration and Informatics (**III**) program;
- Networking Technology and Systems (**NeTS**) program;
- Robust Intelligence (**RI**) program; and
- Software and Hardware Foundations (**SHF**) program.

The acronym should be followed by a colon, then the title of your project. For example, if you are submitting a proposal to the CSR program, then your title would be CSR: Title.

If you submit a proposal as part of a set of collaborative proposals, the words "Collaborative Research" followed by a colon should appear at the beginning of the title, before the program acronym. For example, if you are submitting a collaborative project to the RI program, the title of each proposal would be Collaborative Research: RI: Title. Please note that if submitting via Research.gov, the system will automatically insert the prepended title "Collaborative Research" when the collaborative set of proposals is created.

Proposals from PIs in institutions that have RUI (Research in Undergraduate Institutions) eligibility should have a proposal title that begins with the program acronym followed by a colon, then "RUI" followed by a colon, and then the title, for example, SHF: RUI: Title.

Proposals that extend beyond the scope of one Future CoRe program or area are welcome. In such cases, PIs should identify the acronym for the most relevant program or area, followed by any other relevant program acronym(s) separated by colons, for example, NeTS: AF: Title. In this example, the proposal would be submitted to the NeTS program and would be considered by both the NeTS and AF programs. CISE Program Officers will work with their NSF colleagues to ensure that these proposals are appropriately reviewed and considered for funding.

Additional Supplementary Documents:

In the Other Supplementary Documents section, upload a document with the following information:

A list of Project Personnel and Partner Organizations (required) (Note: In collaborative proposals, the lead organization should provide this information for all participants):

Provide current, accurate information for all personnel and organizations involved in the project. NSF staff will use this information in the merit review process to manage reviewer selection. The list must include all PIs, co-PIs, Senior/Key Personnel, funded/unfunded Consultants or Collaborators, Subawardees, Postdocs, and project-level advisory committee members. This list should be numbered and include (in this order) Full name, Organization(s), and Role in the project, with each item separated by a semi-colon. Each person listed should start a new numbered line. For example:

1. Mei Lin; XYZ University; PI
2. Jak Jabes; University of PQR; Senior/Key Personnel
3. Jane Brown; XYZ University; Postdoctoral Researcher
4. Rakel Ademas; ABC Inc.; Funded Consultant
5. Maria Wan; Welldone Institution; Unfunded Collaborator
6. Rimon Greene; ZZZ University; Subawardee

B. Budgetary Information

Cost Sharing:

Inclusion of voluntary committed cost sharing is prohibited.

C. Due Dates

- **Full Proposal Target Date(s):**

September 11, 2025

Second Thursday in September, Annually Thereafter

February 05, 2026

First Thursday in February, Annually Thereafter

While proposals are accepted anytime, proposers are highly encouraged to submit by the target dates to ensure consideration during the corresponding panel review cycle.

D. Research.gov/Grants.gov Requirements

For Proposals Submitted Via Research.gov:

To prepare and submit a proposal via Research.gov, see detailed technical instructions available at: https://www.research.gov/research-portal/appmanager/base/desktop?_nfpb=true&_pageLabel=research_node_display&_nodePath=/researchGov/Service/Desktop/ProposalPreparationandSubmission.html. For Research.gov user support, call the Research.gov Help Desk at 1-800-381-1532 or e-mail rgov@nsf.gov. The Research.gov Help Desk answers general technical questions related to the use of the Research.gov system. Specific questions related to this program solicitation should be referred to the NSF program staff contact(s) listed in Section VIII of this funding opportunity.

For Proposals Submitted Via Grants.gov:

Before using Grants.gov for the first time, each organization must register to create an institutional profile. Once registered, the applicant's organization can then apply for any federal grant on the Grants.gov website. Comprehensive information about using Grants.gov is available on the Grants.gov Applicant Resources webpage: <https://www.grants.gov/applicants>. In addition, the NSF Grants.gov Application Guide (see link in Section V.A) provides instructions regarding the technical preparation of proposals via Grants.gov. For Grants.gov user support, contact the Grants.gov Contact Center at 1-800-518-4726 or by email: support@grants.gov. The Grants.gov Contact Center answers general technical questions related to the use of Grants.gov. Specific questions related to this program solicitation should be referred to the NSF program staff contact(s) listed in Section VIII of this solicitation.

Submitting the Proposal: Once all documents have been completed, the Authorized Organizational Representative (AOR) must submit the application to Grants.gov and verify the desired funding opportunity and agency to which the application is submitted. The AOR must then sign and submit the application to Grants.gov. The completed application will be transferred to Research.gov for further processing.

The NSF [Grants.gov Proposal Processing in Research.gov informational page](#) provides submission guidance to applicants and links to helpful resources including the NSF [Grants.gov Application Guide](#), [Grants.gov Proposal Processing in Research.gov how-to guide](#), and [Grants.gov Submitted Proposals Frequently Asked Questions](#). Grants.gov proposals must pass all NSF pre-check and post-check validations in order to be accepted by Research.gov at NSF.

When submitting via Grants.gov, NSF strongly recommends applicants initiate proposal submission at least five business days in advance of a deadline to allow adequate time to address NSF compliance errors and resubmissions by 5:00 p.m. submitting organization's local time on the deadline. Please note that some errors cannot be corrected in Grants.gov. Once a proposal passes pre-checks but fails any post-check, an applicant can only correct and submit the in-progress proposal in Research.gov.

Proposers that submitted via Research.gov may use Research.gov to verify the status of their submission to NSF. For proposers that submitted via Grants.gov, until an application has been received and validated by NSF, the Authorized Organizational Representative may check the status of an application on Grants.gov. After proposers have received an e-mail notification from NSF, Research.gov should be used to check the status of an application.

VI. NSF Proposal Processing And Review Procedures

Proposals received by NSF are assigned to the appropriate NSF program for acknowledgement and, if they meet NSF requirements, for review. All proposals are carefully reviewed by a scientist, engineer, or educator serving as an NSF Program Officer, and usually by three to ten other persons outside NSF either as *ad hoc* reviewers, panelists, or both, who are experts in the particular fields represented by the proposal. These reviewers are selected by Program Officers

charged with oversight of the review process. Proposers are invited to suggest names of persons they believe are especially well qualified to review the proposal and/or persons they would prefer not review the proposal. These suggestions may serve as one source in the reviewer selection process at the Program Officer's discretion. Submission of such names, however, is optional. Care is taken to ensure that reviewers have no conflicts of interest with the proposal. In addition, Program Officers may obtain comments from site visits before recommending final action on proposals. Senior NSF staff further review recommendations for awards. A flowchart that depicts the entire NSF proposal and award process (and associated timeline) is included in PAPPG Exhibit III-1.

A comprehensive description of the Foundation's merit review process is available on the NSF website at:

https://www.nsf.gov/bfa/dias/policy/merit_review/.

Proposers should also be aware of core strategies that are essential to the fulfillment of NSF's mission, as articulated in [*Leading the World in Discovery and Innovation, STEM Talent Development and the Delivery of Benefits from Research - NSF Strategic Plan for Fiscal Years \(FY\) 2022 - 2026*](#). These strategies are integrated in the program planning and implementation process, of which proposal review is one part. NSF's mission is particularly well-implemented through the integration of research and education and broadening participation in NSF programs, projects, and activities.

One of the strategic objectives in support of NSF's mission is to foster integration of research and education through the programs, projects, and activities it supports at academic and research institutions. These institutions must recruit, train, and prepare a diverse STEM workforce to advance the frontiers of science and participate in the U.S. technology-based economy. NSF's contribution to the national innovation ecosystem is to provide cutting-edge research under the guidance of the Nation's most creative scientists and engineers. NSF also supports development of a strong science, technology, engineering, and mathematics (STEM) workforce by investing in building the knowledge that informs improvements in STEM teaching and learning.

NSF's mission calls for the broadening of opportunities and expanding participation of groups, institutions, and geographic regions that are underrepresented in STEM disciplines, which is essential to the health and vitality of science and engineering. NSF is committed to this principle of diversity and deems it central to the programs, projects, and activities it considers and supports.

A. Merit Review Principles and Criteria

The National Science Foundation strives to invest in a robust and diverse portfolio of projects that creates new knowledge and enables breakthroughs in understanding across all areas of science and engineering research and education. To identify which projects to support, NSF relies on a merit review process that incorporates consideration of both the technical aspects of a proposed project and its potential to contribute more broadly to advancing NSF's mission "to promote the progress of science; to advance the national health, prosperity, and welfare; to secure the national defense; and for other purposes." NSF makes every effort to conduct a fair, competitive, transparent merit review process for the selection of projects.

1. Merit Review Principles

These principles are to be given due diligence by PIs and organizations when preparing proposals and managing projects, by reviewers when reading and evaluating proposals, and by NSF program staff when determining whether or not to recommend proposals for funding and while overseeing awards. Given that NSF is the primary federal agency charged with nurturing and supporting excellence in basic research and education, the following three principles apply:

- All NSF projects should be of the highest quality and have the potential to advance, if not transform, the frontiers of knowledge.
- NSF projects, in the aggregate, should contribute more broadly to achieving societal goals. These "Broader Impacts" may be accomplished through the research itself, through activities that are directly related to specific research projects, or through activities that are supported by, but are complementary to, the project. The project activities may be based on previously established and/or innovative methods and approaches, but in either case must be well justified.

- Meaningful assessment and evaluation of NSF funded projects should be based on appropriate metrics, keeping in mind the likely correlation between the effect of broader impacts and the resources provided to implement projects. If the size of the activity is limited, evaluation of that activity in isolation is not likely to be meaningful. Thus, assessing the effectiveness of these activities may best be done at a higher, more aggregated, level than the individual project.

With respect to the third principle, even if assessment of Broader Impacts outcomes for particular projects is done at an aggregated level, PIs are expected to be accountable for carrying out the activities described in the funded project. Thus, individual projects should include clearly stated goals, specific descriptions of the activities that the PI intends to do, and a plan in place to document the outputs of those activities.

These three merit review principles provide the basis for the merit review criteria, as well as a context within which the users of the criteria can better understand their intent.

2. Merit Review Criteria

All NSF proposals are evaluated through use of the two National Science Board approved merit review criteria. In some instances, however, NSF will employ additional criteria as required to highlight the specific objectives of certain programs and activities.

The two merit review criteria are listed below. **Both** criteria are to be given **full consideration** during the review and decision-making processes; each criterion is necessary but neither, by itself, is sufficient. Therefore, proposers must fully address both criteria. (PAPPG Chapter II.D.2.d(i). contains additional information for use by proposers in development of the Project Description section of the proposal). Reviewers are strongly encouraged to review the criteria, including PAPPG Chapter II.D.2.d(i), prior to the review of a proposal.

When evaluating NSF proposals, reviewers will be asked to consider what the proposers want to do, why they want to do it, how they plan to do it, how they will know if they succeed, and what benefits could accrue if the project is successful. These issues apply both to the technical aspects of the proposal and the way in which the project may make broader contributions. To that end, reviewers will be asked to evaluate all proposals against two criteria:

- **Intellectual Merit:** The Intellectual Merit criterion encompasses the potential to advance knowledge; and
- **Broader Impacts:** The Broader Impacts criterion encompasses the potential to benefit society and contribute to the achievement of specific, desired societal outcomes.

The following elements should be considered in the review for both criteria:

1. What is the potential for the proposed activity to
 - a. Advance knowledge and understanding within its own field or across different fields (Intellectual Merit); and
 - b. Benefit society or advance desired societal outcomes (Broader Impacts)?
2. To what extent do the proposed activities suggest and explore creative, original, or potentially transformative concepts?
3. Is the plan for carrying out the proposed activities well-reasoned, well-organized, and based on a sound rationale? Does the plan incorporate a mechanism to assess success?
4. How well qualified is the individual, team, or organization to conduct the proposed activities?
5. Are there adequate resources available to the PI (either at the home organization or through collaborations) to carry out the proposed activities?

Broader impacts may be accomplished through the research itself, through the activities that are directly related to specific research projects, or through activities that are supported by, but are complementary to, the project. NSF values the advancement of scientific knowledge and activities that contribute to achievement of societally relevant outcomes. Such outcomes include, but are not limited to: full participation of women, persons with disabilities, and

underrepresented minorities in science, technology, engineering, and mathematics (STEM); improved STEM education and educator development at any level; increased public scientific literacy and public engagement with science and technology; improved well-being of individuals in society; development of a diverse, globally competitive STEM workforce; increased partnerships between academia, industry, and others; improved national security; increased economic competitiveness of the United States; and enhanced infrastructure for research and education.

Proposers are reminded that reviewers will also be asked to review the Data Management and Sharing Plan, Mentoring Plan, and the optional Broadening Participation in Computing Plan, as appropriate.

B. Review and Selection Process

Proposals submitted in response to this program solicitation will be reviewed by

Ad hoc Review and/or Panel Review.

Reviewers will be asked to evaluate proposals using two National Science Board approved merit review criteria and, if applicable, additional program specific criteria. A summary rating and accompanying narrative will generally be completed and submitted by each reviewer and/or panel. The Program Officer assigned to manage the proposal's review will consider the advice of reviewers and will formulate a recommendation.

After scientific, technical and programmatic review and consideration of appropriate factors, the NSF Program Officer recommends to the cognizant Division Director whether the proposal should be declined or recommended for award. NSF strives to be able to tell proposers whether their proposals have been declined or recommended for funding within six months. Large or particularly complex proposals or proposals from new recipients may require additional review and processing time. The time interval begins on the deadline or target date, or receipt date, whichever is later. The interval ends when the Division Director acts upon the Program Officer's recommendation.

After programmatic approval has been obtained, the proposals recommended for funding will be forwarded to the Division of Grants and Agreements or the Division of Acquisition and Cooperative Support for review of business, financial, and policy implications. After an administrative review has occurred, Grants and Agreements Officers perform the processing and issuance of a grant or other agreement. Proposers are cautioned that only a Grants and Agreements Officer may make commitments, obligations or awards on behalf of NSF or authorize the expenditure of funds. No commitment on the part of NSF should be inferred from technical or budgetary discussions with a NSF Program Officer. A Principal Investigator or organization that makes financial or personnel commitments in the absence of a grant or cooperative agreement signed by the NSF Grants and Agreements Officer does so at their own risk.

Once an award or declination decision has been made, Principal Investigators are provided feedback about their proposals. In all cases, reviews are treated as confidential documents. Verbatim copies of reviews, excluding the names of the reviewers or any reviewer-identifying information, are sent to the Principal Investigator/Project Director by the Program Officer. In addition, the proposer will receive an explanation of the decision to award or decline funding.

VII. Award Administration Information

A. Notification of the Award

Notification of the award is made to *the submitting organization* by an NSF Grants and Agreements Officer. Organizations whose proposals are declined will be advised as promptly as possible by the cognizant NSF Program administering the program. Verbatim copies of reviews, not including the identity of the reviewer, will be provided automatically to the Principal Investigator. (See Section VI.B. for additional information on the review process.)

B. Award Conditions

An NSF award consists of: (1) the award notice, which includes any special provisions applicable to the award and any numbered amendments thereto; (2) the budget, which indicates the amounts, by categories of expense, on which NSF has based its support (or otherwise communicates any specific approvals or disapprovals of proposed expenditures); (3) the proposal referenced in the award notice; (4) the applicable award conditions, such as Grant General Conditions (GC-1)*; and (5) any announcement or other NSF issuance that may be incorporated by reference in the award notice.

Cooperative agreements also are administered in accordance with NSF Cooperative Agreement Financial and Administrative Terms and Conditions (CA-FATC) and the applicable Programmatic Terms and Conditions. NSF awards are electronically signed by an NSF Grants and Agreements Officer and transmitted electronically to the organization via e-mail.

*These documents may be accessed electronically on NSF's Website at https://www.nsf.gov/awards/managing/award_conditions.jsp?org=NSF. Paper copies may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov.

More comprehensive information on NSF Award Conditions and other important information on the administration of NSF awards is contained in the NSF *Proposal & Award Policies & Procedures Guide* (PAPPG) Chapter VII, available electronically on the NSF Website at https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.

Administrative and National Policy Requirements

Build America, Buy America

As expressed in Executive Order 14005, [Ensuring the Future is Made in All of America by All of America's Workers](#) (86 FR 7475), it is the policy of the executive branch to use terms and conditions of Federal financial assistance awards to maximize, consistent with law, the use of goods, products, and materials produced in, and services offered in, the United States.

Consistent with the requirements of the Build America, Buy America Act (Pub. L. 117-58, Division G, Title IX, Subtitle A, November 15, 2021), no funding made available through this funding opportunity may be obligated for infrastructure projects under an award unless all iron, steel, manufactured products, and construction materials used in the project are produced in the United States. For additional information, visit NSF's [Build America, Buy America](#) webpage.

C. Reporting Requirements

For all multi-year grants (including both standard and continuing grants), the Principal Investigator must submit an annual project report to the cognizant Program Officer no later than 90 days prior to the end of the current budget period. (Some programs or awards require submission of more frequent project reports). No later than 120 days following expiration of a grant, the PI also is required to submit a final annual project report, and a project outcomes report for the general public.

Failure to provide the required annual or final annual project reports, or the project outcomes report, will delay NSF review and processing of any future funding increments as well as any pending proposals for all identified PIs and co-PIs on a given award. PIs should examine the formats of the required reports in advance to assure availability of required data.

PIs are required to use NSF's electronic project-reporting system, available through Research.gov, for preparation and submission of annual and final annual project reports. Such reports provide information on accomplishments, project participants (individual and organizational), publications, and other specific products and impacts of the project. Submission of the report via Research.gov constitutes certification by the PI that the contents of the report are accurate and complete. The project outcomes report also must be prepared and submitted using Research.gov. This report serves as a brief summary, prepared specifically for the public, of the nature and outcomes of the project. This report will be posted on the NSF website exactly as it is submitted by the PI.

More comprehensive information on NSF Reporting Requirements and other important information on the administration of NSF awards is contained in the NSF *Proposal & Award Policies & Procedures Guide* (PAPPG) Chapter VII, available electronically on the NSF Website at https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.

VIII. Agency Contacts

Please note that the program contact information is current at the time of publishing. See program website for any updates to the points of contact.

General inquiries regarding this program should be made to:

- AF Program Team, telephone: (703) 292 8910, email: cise-af@nsf.gov
- CIF Program Team, telephone: (703) 292 8910, email: cise-cif@nsf.gov
- CSR Program Team, telephone: (703) 292 8950, email: cise-csr@nsf.gov
- CER Program Team, telephone: (703) 292 8910, email: cise-cer@nsf.gov
- CPS Program Team, telephone: (703) 292 8950, email: cise-cps@nsf.gov
- FET Program Team, telephone: (703) 292 8910, email: cise-fet@nsf.gov
- HCC Program Team, telephone: (703) 292 8930, email: cise-hcc@nsf.gov
- III Program Team, telephone: (703) 292 8930, email: cise-iii@nsf.gov
- NeTS Program Team, telephone: (703) 292 8950, email: cise-nets@nsf.gov
- RI Program Team, telephone: (703) 292 8930, email: cise-ri@nsf.gov
- SHF Program Team, telephone: (703) 292 8910, email: cise-shf@nsf.gov

For questions related to the use of NSF systems contact:

- NSF Help Desk: 1-800-381-1532
- Research.gov Help Desk e-mail: rgov@nsf.gov

For questions relating to Grants.gov contact:

- Grants.gov Contact Center: If the Authorized Organizational Representatives (AOR) has not received a confirmation message from Grants.gov within 48 hours of submission of application, please contact via telephone: 1-800-518-4726; e-mail: support@grants.gov.

IX. Other Information

The NSF website provides the most comprehensive source of information on NSF Directorates (including contact information), programs and funding opportunities. Use of this website by potential proposers is strongly encouraged. In addition, "NSF Update" is an information-delivery system designed to keep potential proposers and other interested parties apprised of new NSF funding opportunities and publications, important changes in proposal and award policies and procedures, and upcoming NSF [Grants Conferences](#). Subscribers are informed through e-mail or the user's Web browser each time new publications are issued that match their identified interests. "NSF Update" also is available on [NSF's website](#).

Grants.gov provides an additional electronic capability to search for Federal government-wide grant opportunities. NSF funding opportunities may be accessed via this mechanism. Further information on Grants.gov may be obtained at <https://www.grants.gov>.

About The National Science Foundation

The National Science Foundation (NSF) is an independent Federal agency created by the National Science Foundation Act of 1950, as amended (42 USC 1861-75). The Act states the purpose of the NSF is "to promote the progress of science; [and] to advance the national health, prosperity, and welfare by supporting research and education in all fields of science and engineering."

NSF funds research and education in most fields of science and engineering. It does this through grants and cooperative agreements to more than 2,000 colleges, universities, K-12 school systems, businesses, informal science organizations and other research organizations throughout the US. The Foundation accounts for about one-fourth of Federal support to academic institutions for basic research.

NSF receives approximately 55,000 proposals each year for research, education and training projects, of which approximately 11,000 are funded. In addition, the Foundation receives several thousand applications for graduate and postdoctoral fellowships. The agency operates no laboratories itself but does support National Research Centers, user facilities, certain oceanographic vessels and Arctic and Antarctic research stations. The Foundation also supports cooperative research between universities and industry, US participation in international scientific and engineering efforts, and educational activities at every academic level.

Facilitation Awards for Scientists and Engineers with Disabilities (FASSED) provide funding for special assistance or equipment to enable persons with disabilities to work on NSF-supported projects. See the *NSF Proposal & Award Policies & Procedures Guide* Chapter II.F.7 for instructions regarding preparation of these types of proposals.

The National Science Foundation has Telephonic Device for the Deaf (TDD) and Federal Information Relay Service (FIRS) capabilities that enable individuals with hearing impairments to communicate with the Foundation about NSF programs, employment or general information. TDD may be accessed at (703) 292-5090 and (800) 281-8749, FIRS at (800) 877-8339.

The National Science Foundation Information Center may be reached at (703) 292-5111.

• Location:	2415 Eisenhower Avenue, Alexandria, VA 22314
• For General Information (NSF Information Center):	(703) 292-5111
• TDD (for the hearing-impaired):	(703) 292-5090
• To Order Publications or Forms:	
Send an e-mail to:	nsfpubs@nsf.gov
or telephone:	(703) 292-8134
• To Locate NSF Employees:	(703) 292-5111

Privacy Act And Public Burden Statements

The information requested on proposal forms and project reports is solicited under the authority of the National Science Foundation Act of 1950, as amended. The information on proposal forms will be used in connection with the selection of qualified proposals; and project reports submitted by proposers will be used for program evaluation and reporting within the Executive Branch and to Congress. The information requested may be disclosed to qualified reviewers and staff assistants as part of the proposal review process; to proposer institutions/recipients to provide or obtain data regarding the proposal review process, award decisions, or the administration of awards; to government contractors, experts, volunteers and researchers and educators as necessary to complete assigned work; to other government agencies or other entities needing information regarding proposers or nominees as part of a joint application review process, or in order to coordinate programs or policy; and to another Federal agency, court, or party in a court or Federal administrative proceeding if the government is a party. Information about Principal Investigators may be added to the Reviewer file and

used to select potential candidates to serve as peer reviewers or advisory committee members. See [System of Record Notices](#), [NSF-50](#), "Principal Investigator/Proposal File and Associated Records," and [NSF-51](#), "Reviewer/Proposal File and Associated Records." Submission of the information is voluntary. Failure to provide full and complete information, however, may reduce the possibility of receiving an award.

An agency may not conduct or sponsor, and a person is not required to respond to, an information collection unless it displays a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 3145-0058. Public reporting burden for this collection of information is estimated to average 120 hours per response, including the time for reviewing instructions. Send comments regarding the burden estimate and any other aspect of this collection of information, including suggestions for reducing this burden, to:

Suzanne H. Plimpton
Reports Clearance Officer
Policy Office, Division of Institution and Award Support
Office of Budget, Finance, and Award Management
National Science Foundation
Alexandria, VA 22314

[Vulnerability disclosure](#) | [Inspector General](#) | [Privacy](#) | [FOIA](#) | [No FEAR Act](#) | [USA.gov](#) | [Accessibility](#) | [Plain language](#) |



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